

SURFACE

OCEAN

WANDERERS, such as tropicbirds, spend most of their lives far out at sea, returning to land only to breed.

MIDWATER

THE HUGE OCEAN SUNFISH

is a typical open water, or "pelagic" species: it spends its life often hundreds of miles from land.

MARINE TURTLES

often migrate long distances between their feeding grounds and the beaches where they breed.

ABYSSAL ZONE

DOLPHINS have streamlined bodies that give them exceptional swimming ability. They often roam far out of sight of land.

THE VIPERFISH is one of many deep-sea species that produce light in order to attract their prey.

SEABED

DEEP-SEA SCAVENGERS, such as brittlestars, feed on organic matter that drifts down to the seabed.

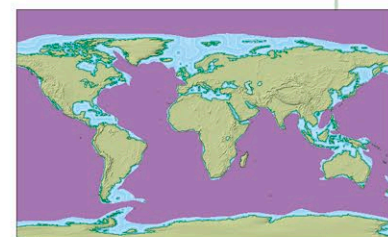
Open sea

Even in the clearest sea water, light penetrates no farther than about 825 ft (250 m) below the surface. More than any other factor, this has a crucial effect on sea life because it determines what there is to eat. In the brightly lit surface zone, microscopic

algae grow by harnessing the energy in sunlight, creating an invisible harvest for planktonic animals. This food is passed on when the plankton are themselves eaten in food chains that eventually end with sharks and other large predators. Below the point where the light fades and finally disappears, there is no home-grown supply of food. At this depth, and further below, animals feed either on each other or on the supply of dead remains that constantly drifts down from the surface.

Despite the unimaginable volume of the oceans, few of the world's animal species—perhaps 5 percent—live in open water. Since most of these stay near the surface, where they can take advantage of the relatively plentiful food supply, animal life in the huge midwater zone and the deeper abyssal zone is comparatively sparse.

In contrast with these central zones, much of the ocean floor abounds with animals. Those that live on the seabed are called benthic animals and include species that swim or crawl over the seabed as well as those that burrow through the soft sediment for food. Many appear to have changed little over millions of years, for although the water is very cold, and the pressure intense, the deep seabed is not subjected to the changeable conditions that can affect the surface. It is thus one of the most stable habitats on earth.



IF THE CONTINENTAL shelves are excluded, the average depth of the world's ocean basins is about 13,200 ft (4,000 m).

OVERFISHING

The world's total fish catch currently stands at nearly 103 million tons (93.4 million tonnes) a year, with about nine-tenths coming from the sea. This colossal harvest has a major impact on marine ecosystems, with fishing spreading from traditional grounds to polar waters and the deep seabed. Managed appropriately, marine fish can be sustainably harvested, whereas overfishing can disrupt breeding, pushing stocks to the point of collapse. It is estimated that close to 30 percent of the world's

commercial fish stocks are overfished and only 10 percent are classed as underfished.



SURE CATCH

Using sonar and global positioning systems, modern trawlers can track down schools with pinpoint accuracy. Without strict controls, far fewer adult fish are left to breed.